

## Question 1

The umbilical cord links a baby to the mother and carries nutrients and gases for the growth of baby. In the past a huge portion of the baby cord was discarded. Now it is believed that the umbilical cord blood stem cells can possibly help cure several diseases. As such, many individuals now store their umbilical cord after giving birth. The cord is rich in two types of stem cells, they are hematopoietic stem cells and mesenchymal cells.

(a) Describe two features of stems cells that normal adult cells do not possess. [2]

- **The ability to self-renew**
- **The ability to differentiate.**
- **They are unspecialised**

(b) The two types of stem cells found in the umbilical cord are considered by science to be multipotent. Explain what this means and suggest how such stem cells could potentially be used to cure human diseases without involving gene therapy. [3]

- **Able to differentiate into some adult cell types**
- **Suitable suggestion for curing human diseases. [2]**

Sometimes, DNA from stem cells is extracted for use in further medical research. The amounts extracted are usually very small and is seldom enough for practical use. A technique that can be employed to increase the amounts of extracted DNA is the Polymerase Chain Reaction (PCR). Figure 1.1 on the following page illustrates what happens to DNA during this process with a specific focus on the temperatures involved.

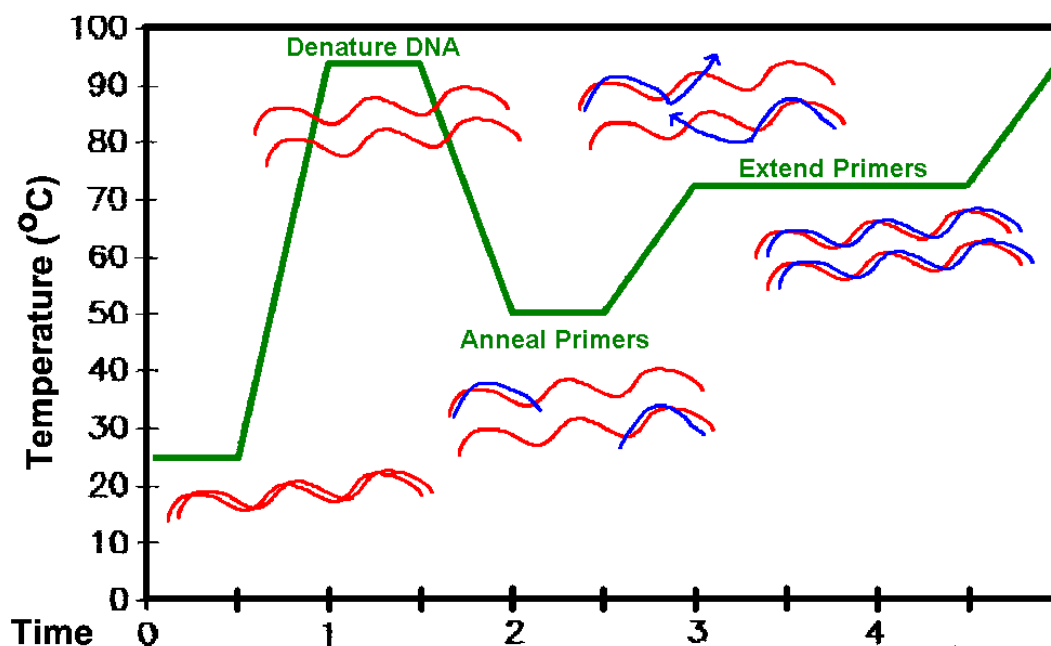


Figure 1.1

(c) Explain the significance of using different temperatures to this process. [3]

- **Allows for denaturation of double stranded DNA into single strands**
- **Which allow for primers to anneal**
- **Further change of temperature can then allow Taq Polymerase to operate optimally.**

(d) Describe the key differences between this process and DNA replication in human cells. [3]

- **DNA primers vs RNA primers**
- **Taq polymerase vs DNA polymerase**
- **Use of heat to denature DNA vs use of helicase to denature DNA**
- **AVP**

(e) This process will not be able to proceed without the use of large amounts of specific primers. Explain why large amounts are needed and why they need to be specific. [2]

- **Large amounts are required to ensure that primers anneal to DNA strands instead of the strands re-annealing with each other.**
- **They need to be specific so that they anneal to the correct region flanking the DNA is amplified via complementary base pairing.**

[Total: 13 marks]

## Question 2

Huntington's disease (HD) is a lethal neurodegenerative disease that is inherited as an autosomal dominant condition. As the appearance of symptoms may occur later in life (as late as at age 40), patients with HD may unknowingly pass on the mutant HD allele to their children.

A radioactive G8 probe revealed two RFLP variants that are linked to the HD allele which can be used in the screening of HD. Figure. 2.1 shows the two RFLP variants upstream to the HD allele and the G8 probe. The arrows show the position of *Hind*III restriction sites. The numbers represent kilobases.

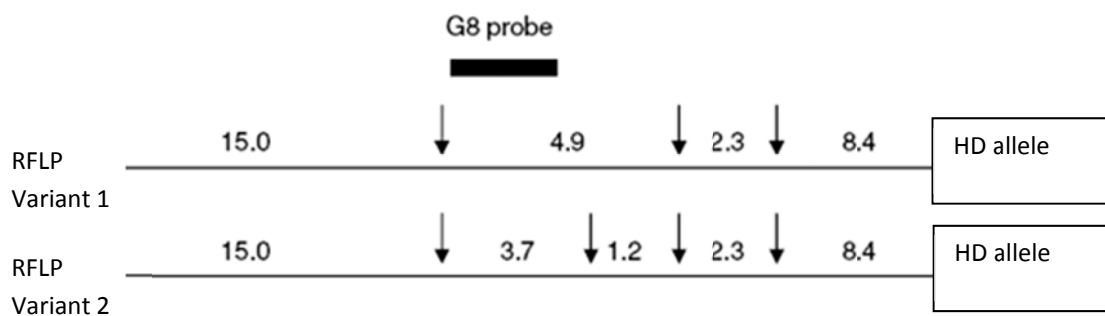


Figure. 2.1

Figure. 2.2 illustrates the Southern Blot analysis of the two RFLP variants in 30 related individuals with a family history of HD.

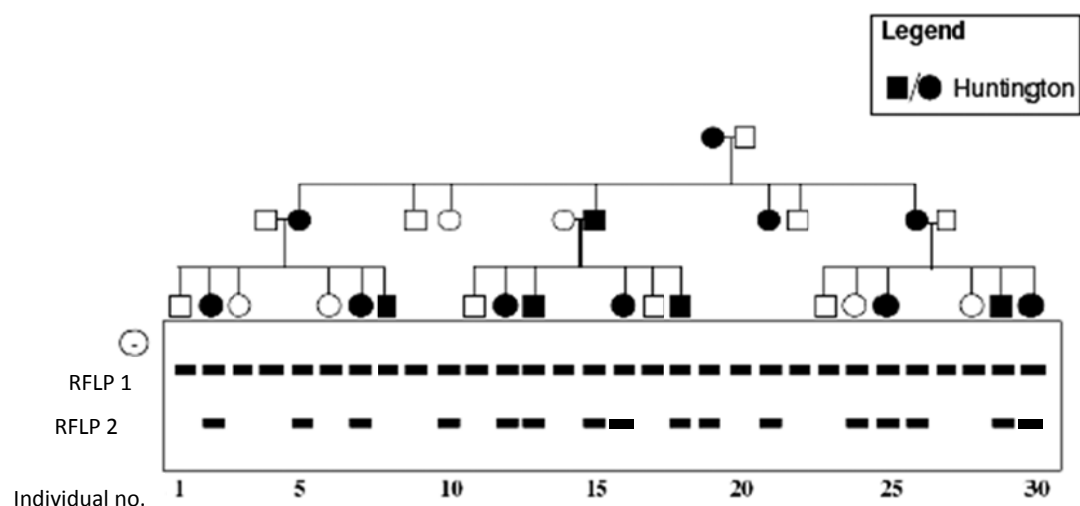


Figure. 2.2

With reference to Fig. 2.1 and 2.2,

(a) outline how the DNA band patterns of the 30 individuals were obtained. [4]

1. DNA sample from the 30 individuals is treated/digested with *HindIII* (into restriction fragments  
Restriction fragments are subjected to gel electrophoresis where the fragments are separated according to sizes.
2. Shorter fragments move further OR cite figures (eg. 1.2kb fragments move further than the 4.9kb fragments)
3. The separated DNA bands are transferred to nitrocellulose membrane by Southern Blotting  
Double stranded DNA denatured by NaOH / alkali
4. Flushed with solution of G8 probe which is complementary to / which binds to / anneals to / hybridizes with / forms hydrogen bonds with 4.9kb and 3.7kb fragments  
Viewed by autoradiography / radioactivity exposes photographic film

(b) identify the RFLP variant which is linked to the mutant HD allele. Explain your answer. [2]

- **RFLP 2**

Explanation:

- Most affected individuals have RFLP variant 2 except individual 8
- Most unaffected individuals do not have RFLP variant 2 except individuals 10, 24
- Most affected individuals have a shorter 3.7kb fragment except individual 8
- Most unaffected individuals have a 4.9kb fragment except individual 24

(c) suggest two possible reasons for the difference in the DNA band patterns among unaffected individuals. [4]

- Individual 10 and 24 received from affected parent a chromosome with mutant HD allele linked to RFLP 1...
- ...although the normal HD allele is linked to RFLP 1

OR

- Mutation in a *HindIII* restriction site of RFLP 1 / in 4.9kb fragment led to...
- ... RFLP1 becoming RFLP2 (OWTTE) / appearance of one more *HindIII* restriction site. Hence, variant 1 not detected in individuals 10 & 24.

OR

- During the formation of ovum in the female parent of individual 10 and 24, crossing over occurred at a region between the RFLP locus and HD locus...
- ... cause RFLP2 to become associated with normal allele

OR

- Individual 10 and 24 have the mutant allele associated with RFLP 2 but symptoms have not appeared
- ... as disease occur later in life / as late as 40 years / person still young

(d) discuss the effectiveness of using this RFLP marker in the screening of HD.[2]

- not very effective / may lead to wrong diagnosis
- the linkage between RFLP and the HD allele is not close enough to prevent crossing over from occurring

[Total: 12 marks]

### Question 3

The following Figure 3.1 illustrates the prevalence of **genetically modified crops** in the world over a number of years.

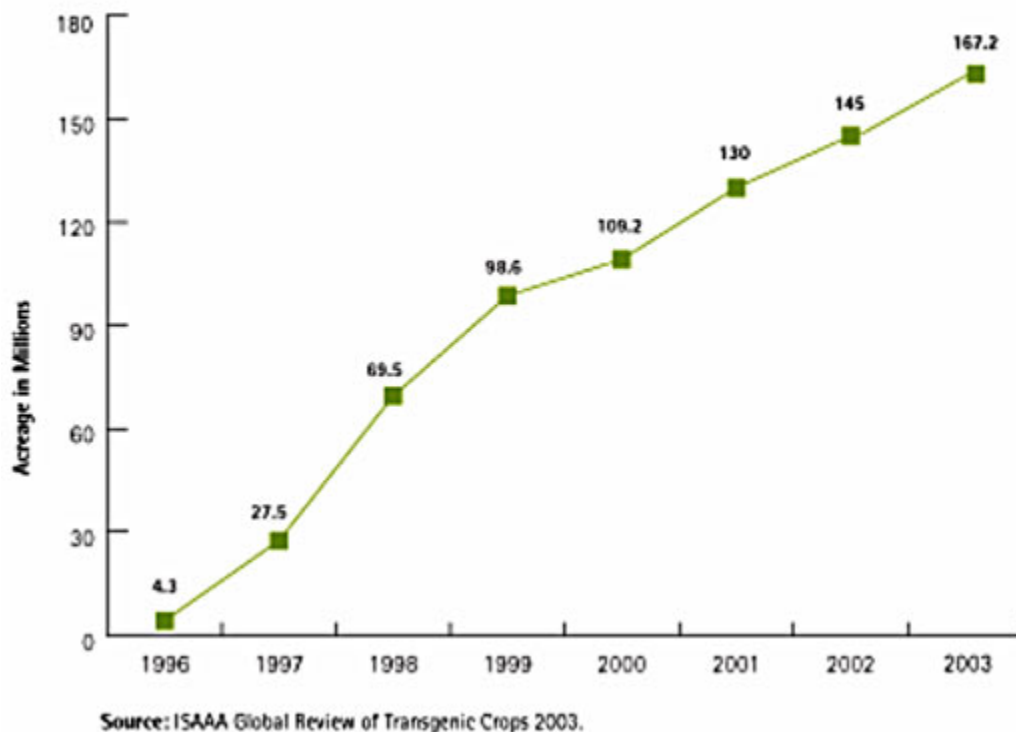


Figure 3.1

(a) Describe what you understand by the term '**genetically modified crops**'. [2]

- **Plants with a foreign gene inserted into all their cells.**
- **The foreign gene is expressed to provide an economic benefit.**

(b) With reference to Figure 3.1, describe the growth in genetically modified crops between the years 1998 and 2003. [2]

- **Acreage of genetically modified crops increases steadily between 1998 to 2003.**
- **Quote data**

**Reject – Answers that's start with 1996**

(c) Explain, with the use of a named example, how genetically modified crops has led to positive outcomes for mankind. [4]

- **Insertion of EPSP synthase gene with an active promoter into crop plants;**
- **leads to increased production of three essential amino acids – tryptophan, tyrosine and phenylalanine, required for protein synthesis**
- **Overcomes the effect of the herbicide Glyphosate (Roundup®) on crop plants**
- **Leading to increased yield of crop plant.**

**OR**

- **Insertion of a gene from the *Streptomyces hygroscopicus* into crop plants that can inactivate glufosinate (A herbicide also called Liberty®)**
- **Glufosinate no longer inhibits the enzyme glutamine synthetase**
- **Glutamine can be produced from ammonia in crop plants exposed to the herbicide.**
- **Leads to increased yield of crop plant.**

**OR**

- **Insertion of a gene from *Bacillus thuringiensis* into crop plants (BT gene)**
- **Leads to the production of Cry Protein.**
- **Which kills insects that eat its leaves.**
- **Leading to increased yield of crop plant.**

**OR**

- **Insertion of a gene comes from the daffodil and another gene from a bacterium found in the soil, *Erwinia uredovora* into rice plants.**
- **rice plants can now produce the enzymes necessary to convert GGDP to provitamin A**
- **Humans can convert Provitamin A into Vitamin A for nutrition.**
- **Leads to increased quality of crop plant.**

(d) In spite of possible positive outcomes, genetically modified crops have yet to be fully accepted by many countries. In fact, many individuals are openly against the use of genetic modification for food products. Comment. [2]

- **Fear of allergens produced due to the genetic modification causing disease/harm in humans.**
- **Fear of antibiotic resistance genes transferring to intestinal bacteria.**
- **Fear of genes introduced via genetic modification escaping to the natural flora thus creating an imbalance in nature (superweeds/superbugs)**

Figure 3.2 illustrates how plant tissue culture techniques may be leveraged upon to bring about genetic modification in plant tissues.

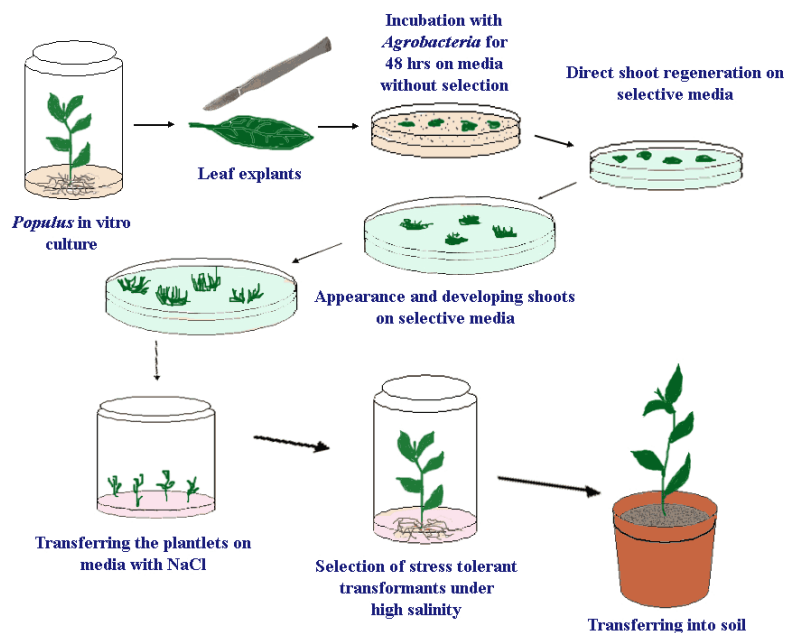


Figure 3.2

(e) Using the information from Figure 3.2, describe the process of cloning plants through plant tissue culture techniques. [3]

- the surfaces of leaf explants are sterilised using dilute sodium hypochlorite and are transferred aseptically to culture plates containing nutrients and plant growth substances.
- The cells of an explant divide by mitosis to form a callus, which is an undifferentiated, actively dividing mass of cells.
- The cells of a callus can be induced to produce shoots and roots by varying the combination of the various plant growth substances.
- The resulting plantlet can be transferred to other media for selection and then eventually transferred to soil.

(f) *Agrobacterium* is commonly used in the above process as depicted in Figure 3.2. Explain the role of *Agrobacterium* in this process. [2]

- *Agrobacterium* can be modified to carry a gene beneficial to mankind.
- *Agrobacterium* is able to insert this gene into the plant cell being cultured.

[Total: 15 marks]

#### Question 4 – Planning Question

In mammals, the products of digestion of food are absorbed from the intestinal cavity.

To investigate the uptake of glucose, a short piece of intestine is turned inside out, so that the absorptive surface is on the outside. The ends of the intestine are tied to form a bag containing a buffer solution, which is then placed in a boiling tube with a known concentration of glucose solution.

The concentration of glucose can be determined by using a colour standard. This is prepared by using a series of glucose solutions of different concentrations.

Using this information and your own knowledge, design an experiment to determine the effect of glucose concentration on the rate of uptake of glucose by the piece of intestine.

You must use:

- 10 % glucose solution
- Distilled water
- Pieces of intestine tied to form a bag containing buffer solution
- Benedict's solution
- 5 glucose solutions at 0.1%, 0.5%, 1%, 2% and 4%
- Isotonic buffer solution

You should select from the following apparatus:

- Normal laboratory glassware e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc
- Timer e.g. stopwatch
- Syringes
- Bunsen burner with tripod, gauze and bench mat
- Thermometer

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams if necessary,
- identify the dependent and independent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment

## MARKS SCHEME

### 1. Theoretical consideration [3 marks max]:

- ✓ **Theory and explanation:** (max 1)
  - o Intestinal membrane is embedded with **transport proteins/ transmembrane proteins/ carrier proteins**;
  - o allowing the entry of glucose into the intestine bag via **facilitated diffusion**.
- ✓ **Experimental Design:** (max 1)
  - o The **concentration of glucose** present in the intestine bag over a **fixed duration of time** can be determined by performing **Benedict's test** on the solution in the boiling tube;
  - o And **compared** this test results with the **results of a colour standard** which is prepared by using a **series of glucose solutions of different concentrations**.
  - o The **concentration of glucose taken up** by the piece of intestine can be determined by **the concentration of glucose in the intestine bag**.
- ✓ **Hypothesis:** (1M)
  - o Rate of glucose uptake **increases** with an increase in glucose concentration initially and gradually **reaches a constant rate/plateaus** with further increase in glucose concentration.

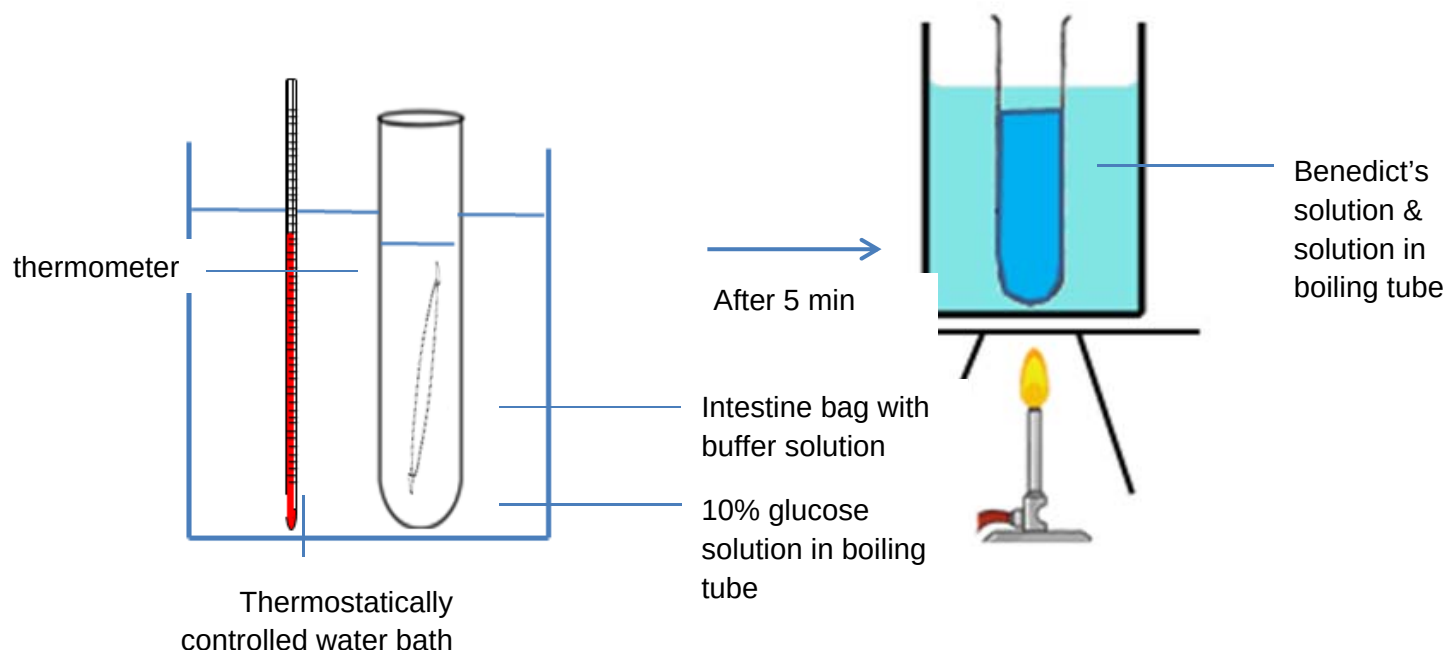
### 2. Variables [2 M for all 3 variables]:

1. **Independent variable:** glucose concentration (**%**); give **at least 5 values**; eg 2.0, 4.0, 6.0, 8.0, 10.0 %
2. **Dependent variable:** Rate of glucose uptake (**% min<sup>-1</sup>**)
3. Other variables to be **kept constant / Controlled variables (Any 2):**
  1. Volume of glucose solution in all tubes
    - o *Ensure a constant of 10cm<sup>3</sup> of glucose solution is used by using a measuring cylinder.*
  2. Duration for testing
    - o *Kept the duration of 5 minute a constant by using a stopwatch.*
  3. Temperature
    - o *Kept the incubation temperature constant by placing the tubes in a thermostatically controlled water bath at 37°C.*

### 3. Control set-up [1 M]:

To prove that the uptake of glucose is due to the presence of the intestine membrane, a **control (test tube)** is set up. **Steps 2 – 14 are performed with all other experimental conditions remained the same** except using **10 cm<sup>3</sup> of distilled water**, in place of glucose solution.

#### 4. Labelled Diagram [1 M]:



#### 5. Procedures [Award only if use of apparatus and quantity stated]

- Using dilution, prepare **5 glucose concentrations** of 2.0%, 4.0%, 6.0%, 8.0% and 10.0%, from the stock of 10.0% glucose. [0.5 mark]:

| Volume of Distilled water (cm <sup>3</sup> ) | Volume of 10% Glucose (stock) Solution (cm <sup>3</sup> ) | Concentration of Glucose Solution (%) |
|----------------------------------------------|-----------------------------------------------------------|---------------------------------------|
| 16                                           | 4                                                         | 2.0                                   |
| 12                                           | 8                                                         | 4.0                                   |
| 8                                            | 12                                                        | 6.0                                   |
| 4                                            | 16                                                        | 8.0                                   |
| 0                                            | 20                                                        | 10.0                                  |

NOTE: total volume of glucose solution is **20 cm<sup>3</sup>** [0.5 mark for dilution table]:

- Obtain the colour standards by first labelling each of the 5 test tubes as **0.5%, 1%, 2% and 4% glucose concentration**.
- Add **2 cm<sup>3</sup> of each of standard glucose solution to each tube** and then add **2 cm<sup>3</sup> of Benedict's solution into the respective tubes**. Shake to mix well the contents and boil the mixture for 5 minutes.
- After 5 minutes, observe the contents of each of tubes by noting the colour of the precipitate formed.
- Place the **intestine bag of buffer solution** into a **boiling tube of isotonic buffer**. This is to **prevent the intestine from drying up and also to maintain its osmotic pressure**.
- Using a **syringe**, add **10 cm<sup>3</sup> of 10% glucose solution into a boiling tube**.

7. Incubate both boiling tubes in the **thermostatically controlled water bath at 37 °C** for **5 min** to allow the **temperature to equilibrate**. Monitor temperature using **thermometer**.
8. Use a **stopwatch** to monitor equilibration time.
9. **At end of 5 minutes**, place the intestine bag of buffer solution into the boiling tube with 10 cm<sup>3</sup> of 10% glucose solution, and incubate at **37 °C** for **5 min**.
10. After **5 minutes**, remove the intestine bag and transfer the glucose solution into a beaker. Use a **syringe** to remove **2 cm<sup>3</sup> of glucose solution** and place it inside a test-tube.
11. Perform the **Benedict's test** by adding **2 cm<sup>3</sup> of Benedict's solution** into the test-tube containing **2 cm<sup>3</sup> of glucose solution**. Shake to mix well the contents and boil the mixture for 5 minutes.
12. **Compare** the result of this test with those of the colour standards obtained earlier and **record the estimated concentration of glucose concentration**.
13. To ensure reproducibility of results, **perform 2 more replicates** by repeating steps 2 to 12.
14. To ensure reliability of data, **repeat the entire experiment** (steps 1 to 12) twice.

#### 6. Manipulation of data [0.5 M each]

- ✓ Calculate the **average concentration of glucose solution** for the 3 replicates;
- ✓ The **rate of glucose taken up by** the piece of intestine can then be calculated as **1/concentration of glucose in intestine bag (% min<sup>-1</sup>)**.

#### 7. Table and Data [1 M]:

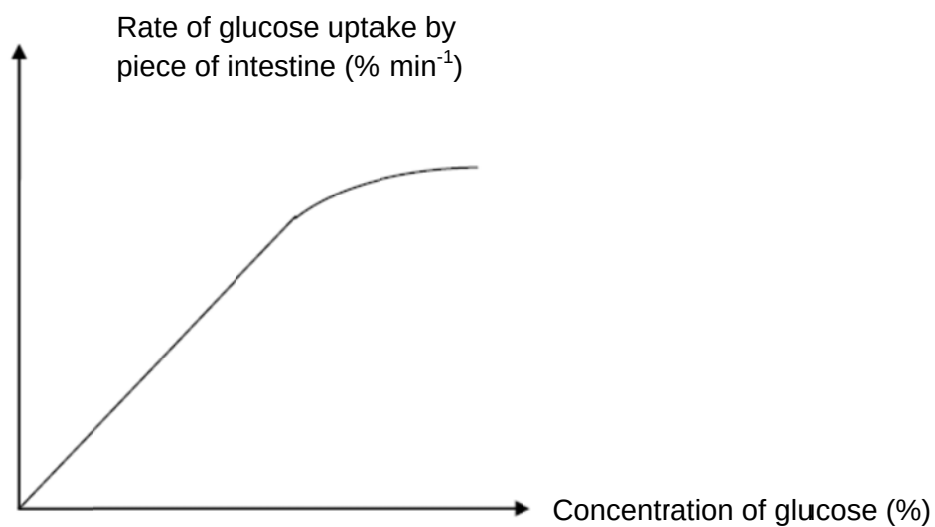
- ✓ **Table showing the rate of glucose uptake by piece of intestine at different glucose concentration**

| Initial glucose concentration (%) | Concentration of glucose solution inside intestine bag (%) |           |           |         | Rate of glucose taken up by piece of intestine (% min <sup>-1</sup> ) |
|-----------------------------------|------------------------------------------------------------|-----------|-----------|---------|-----------------------------------------------------------------------|
|                                   | Reading 1                                                  | Reading 2 | Reading 3 | Average |                                                                       |
| 2.0                               |                                                            |           |           |         |                                                                       |
| 4.0                               |                                                            |           |           |         |                                                                       |
| 6.0                               |                                                            |           |           |         |                                                                       |
| 8.0                               |                                                            |           |           |         |                                                                       |
| 10.0                              |                                                            |           |           |         |                                                                       |

#### 8. Graph [1 M]:

- ✓ **Labeled axes and correct proposed trend**

**Graph of rate of glucose uptake by piece of intestine ( $\% \text{ min}^{-1}$ ) against glucose concentration (%).**



**9. Risks & Precautions [1M]:**

1. Mercury is **toxic** and can lead to **poisoning**. Exercise caution in handling thermometer so as to prevent breakage.
2. Benedict's solution is a **skin irritant**. Wear gloves when handling the reagent.

### Question 5 - Free-response question

- (a) Describe the genetic basis of cystic fibrosis and compare viral and non-viral delivery systems in the treatment of cystic fibrosis. [8]

#### Genetic basis of CF:

1. Genetic disease due to a deletion mutation of 3 nucleotides of CFTR gene on chromosome 7, resulting in the loss of phenylalanine in polypeptide;
2. cystic fibrosis transmembrane conductance regulator (CFTR) becomes non-functional;
3. CF is an autosomal recessive inherited disorder

Viral delivery systems for treatment of cystic fibrosis involves the use of adenoviruses and adeno-associated viruses; Non-viral delivery system for treatment of cystic fibrosis involves use of liposomes OR gene gun.

From RI 2013 Prelims

|     | Point of comparison                                    | Adenovirus-mediated gene therapy                                                                                                                             | Liposome-mediated gene therapy                                                                                                                                            |
|-----|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <b>Differences</b>                                     |                                                                                                                                                              |                                                                                                                                                                           |
| D1  | <u>Immune response</u>                                 | Adenovirus more likely to <u>evoke immune response</u>                                                                                                       | Less likely to <u>evoke immune response</u>                                                                                                                               |
| D2  | <u>Effectiveness on evoking immune response</u>        | <u>Less likely to be effective</u> after the body has <u>developed immunity</u> to the viral vector                                                          | <u>Remains effective</u> as body <u>does not develop immunity</u> to the vector                                                                                           |
| D3  | <u>Disease causing</u> / risk assessment               | Adenoviral vector (which does not integrate into the genome can cause throat <u>infections</u> but not human malignancies (cancer)                           | <u>Do not cause disease at all / safer</u> (Liposomes have a small degree of toxicity – so many rounds of treatment can be harmful to the cells or tissues being treated) |
| D4  | Specificity                                            | More <u>specific targeting</u> of cells                                                                                                                      | Less <u>specific targeting</u> of cells                                                                                                                                   |
| D5  | Transfection efficiency                                | <u>High</u> transfection efficiency.                                                                                                                         | <u>Low</u> transfection efficiency.                                                                                                                                       |
| D6  | Delivery of normal functional CFTR allele into nucleus | Greater chances of delivering normal functional allele into nucleus via nuclear pore as adenovirus directly injects DNA at nuclear pore.                     | No ability to transfer normal functional allele into nucleus.                                                                                                             |
| D10 | Entry into cell                                        | Virus enters cell via <u>endocytosis</u> , delivering normal functional CFTR allele into target cell.                                                        | <u>Fusion of lipid bilayer</u> of liposome with cell membrane leads to delivery of normal functional CFTR allele into cell.                                               |
|     | <b>Similarities</b>                                    |                                                                                                                                                              |                                                                                                                                                                           |
| S1  | Integration                                            | Liposomes and adenoviruses not able to <u>integrate normal functional CFTR allele into genome</u> (extrachromosomal) / do not cause insertional mutagenesis; |                                                                                                                                                                           |

|    |                                          |                                                                                                                                                                                                                                 |
|----|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                                          | And hence the treatment is short-lived;                                                                                                                                                                                         |
| S2 | Duration of therapeutic effects          | Treatment is <u>transient</u> as <u>epithelial cells are constantly being shed</u> , so repeated therapy is needed;                                                                                                             |
| S3 | Target organs                            | Cystic fibrosis <u>affects other organs</u> that are less accessible e.g., <u>pancreas</u> and <u>small intestines</u> , and the epithelial cells of the pancreas is a challenge to target by both liposomal or viral delivery; |
| S4 | Target cells                             | Difficult to target the stem cells of the epithelium of every organ                                                                                                                                                             |
| S5 | <i>In vivo</i> / <i>ex vivo</i> approach | Both systems are carried out using an <i>in vivo</i> approach because it is difficult to extract epithelial cells from respiratory tract.                                                                                       |
| S6 | Level of expression                      | <u>Fine-tuning</u> the expression of normal functional CFTR allele within target cells is <u>difficult</u> ; under-expression of normal functional CFTR allele in target cells would render gene therapy ineffective.           |

|    | Point of comparison                                           | Retroviral-mediated gene therapy                                                                                                                                              | Liposome-mediated gene therapy                                            |
|----|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| D7 | Insertional mutagenesis                                       | Retroviral vector may induce a tumour via <u>insertional mutagenesis</u> of <u>e.g. tumour suppressor gene</u> by <u>integrating normal functional allele into the genome</u> | No insertional mutagenesis.                                               |
| D8 | Integration of normal functional CFTR allele into host genome | Retrovirus possesses integrase which integrates normal functional CFTR allele into host genome.                                                                               | Absence of integration of normal functional CFTR allele into host genome. |
| D9 | Stability of gene expression                                  | <u>Stable</u> gene expression.                                                                                                                                                | <u>Unstable</u> and inconsistent gene expression.                         |

(b) Discuss the social and ethical considerations for the use of gene therapy. [6]

### Issues on the safety and efficacy of gene therapy

1. Presence of **limitations in techniques/uncertainties of outcomes** as well as **potential clinical risks**, which may lead to **unknown long term consequences** to the well-being of the recipients.
2. It is difficult to determine the **long term effects of exposure to viral vectors** on the human genome.

### Extremely high cost

3. Gene therapy is currently very expensive, require expertise and equipment found only in major medical centres, concerns about gene therapy become a **luxury available only to the rich and powerful**.
4. This could possibly lead to **social disparity between the rich and the poor**, with the rich having better accessibility to treatments.

### Potential impact on improving patients' quality of life

5. Gene therapy might **provide alternative forms of treatments** directed at currently unmet clinical needs, where conventional treatments have failed.
6. **Germline therapy** can eradicate disease from subsequent generation and prevent transmission of serious genetic diseases from parent to offspring

### Potential practice of eugenics

7. (**non-therapeutic enhancement possibilities**), for example, improving memory and intelligence by genetic manipulation
8. concerns about the widespread use of gene therapy making **society less accepting of people** who have mild disorders / genetic diseases or less able;
9. raises ethical issues such **type of circumstances that determines whether genomes should be altered** and the **decision of what is a disability or disorder**.

### Interfere with evolution

10. elimination of unwanted alleles from the gene pool will decrease **genetic variation**;
11. **detrimental to the survival of our species in future**  
/ genes that are damaging under some conditions may be advantageous under other conditions / germ-line gene therapy would forever change the genetic makeup of an individual's descendants / the human gene pool would be permanently affected

### Interfere with life

12. Altering ones' genes and changing the way one was created go against many **religious and cultural beliefs**.
13. The **unborn child** who would be affected by germline gene therapy **cannot choose** whether to have the treatments

(c) Explain the problems associated with the expression of eukaryotic genes in prokaryotes and suggest how these problems can be overcome. [6]

|           | <b>Problem</b>                                                                                                                                                                                           | <b>Solution</b>                                                                                                                                                                                                                                                                                          |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1.</b> | <p>Prokaryotes <b>cannot remove introns</b> and thus <b>cannot convert pre-mRNA to mRNA</b>. [1]</p> <p>This is because bacterial cells <b>do not have RNA splicing machinery/ spliceosomes</b>. [1]</p> | <p>Employ <b>reverse transcription</b>, where <b>mRNA</b> is used as a <b>template</b> by <b>reverse transcriptase</b> to catalyse the synthesis of <b>complementary DNA</b> [1];</p> <p><b>DNA polymerase III</b> then catalyses synthesis of another cDNA, forming <b>double stranded DNA</b> [1].</p> |
| <b>2.</b> | <p><b>RNA polymerases in bacterial cells</b> may not recognize <b>eukaryotic promoter</b>, thus gene is not transcribed. [1]</p>                                                                         | <p>Use an <b>expression vector containing prokaryotic promoter</b>, located upstream of a restriction site where the eukaryotic gene can be inserted. [1]</p> <p><b>Bacterial RNA polymerases</b> recognise <b>prokaryotic promoter</b>, thus transcribing the gene downstream to it. [1]</p>            |
| <b>3.</b> | <p>Lack of <b>post-translational modifications of eukaryotic proteins</b> due to the <b>lack of enzymes</b>. [1]</p>                                                                                     | <p>Use of <b>eukaryotic cell systems</b> like <b>yeasts as host cells</b> [1];</p> <p>Enable <b>glycosylation/ correct folding</b> into the tertiary and quaternary structures, producing <b>functional proteins</b>. [1]</p>                                                                            |